Exam #2

Name ______________________

fsuid ______________________

MAA 4402 JULY 21, 2023

Your exam should have 7 problems plus 1 bonus problem on 10 pages, including scratch paper. The score will be computed out of a total of 100 points. Make sure that you write clearly and neatly on your exam and confine your scratch work to the scratch paper. When applicable, make sure to justify your answers. No calculators or notes may be used on this exam. Finally, make sure to remember to

Stay calm and attempt each problem

Problem	Points	Possible
1		15
2		15
3		14
4		14
5		14
6		14
7		14
Bonus		5
Total		100

1. For each of the following statements indicate whether it is true or false and provide a brief explanation for your answer **(5 points each)**

   a. Let $\mathsf{Log}(z)$ be the principal logarithm. If $w \in \mathbb{C}$ then $\mathsf{Log}(w^2) = 2\mathsf{Log}(w)$

   b. If $w, z \in \mathbb{C}$ and $\mathsf{Re}(w), \mathsf{Re}(z) > 0$, then $\mathsf{Log}(wz) = \mathsf{Log}(w) + \mathsf{Log}(z)$

   c. Let $f(z)$ be a complex valued function defined on $U \subset \mathbb{C}$ and let γ and $\tilde{\gamma}$ be two curves in U connecting points z_1 and z_2, then $\int_\gamma f(z)dz = \int_{\tilde{\gamma}} f(z)dz$.

2. Below you will be given several functions $f(z)$ and complex numbers c. Compute $f(c)$ and $f'(c)$. If $f(z)$ is multivalued make sure to compute all values of $f(c)$ and $f'(c)$. **(5 points each)**

   a. $f(z) = z^{\frac{1}{2}+i}$, $c = 1$

   b. $f(z) = \sin z$, $c = 2i$.

   c. $f(z) = \log z$, $c = \frac{i}{\pi}$

3. Let $f(z) = ze^{\overline{z}}$.

   a. Compute $\frac{\partial f}{\partial \overline{z}}$ **(7 points)**
   Hint: use the product rule

   b. Find all points where $f(z)$ is differentiable and compute the derivative at these values. **(7 points)**

4. Let $f(z) = z\cos(z^2)$ and let γ be the top half of the circle centered at $\pi/2$ of radius $\pi/2$, traversed counterclockwise

   a. Sketch γ. **(7 points)**

   b. Compute $\int_\gamma f(z)dz$. **(7 points)**

5. Let γ be the top half of the unit circle (traversed counterclockwise). γ can be parameterized by $\gamma(t) = e^{it}$, $0 < t < \pi$. Compute **(14 points)**

$$\int_\gamma \log(z)dz$$

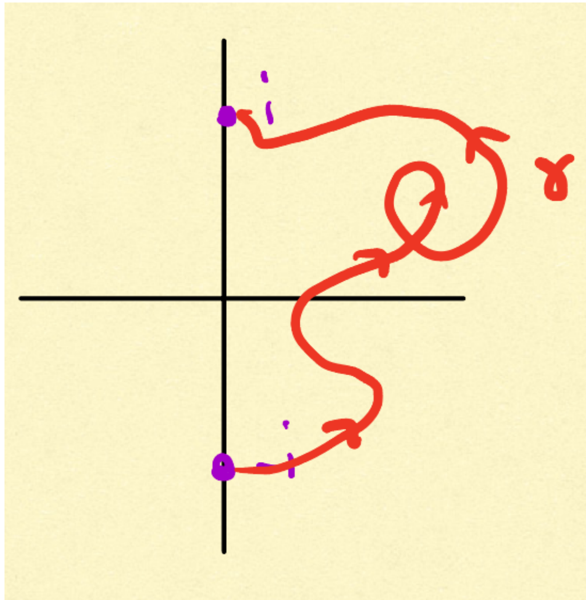
using the branch $F(z) = \ln(|z|) + i\theta$, where $z = |z|\, e^{i\theta}$ and $0 < \theta < 2\pi$.

6. Let $f(z)$ be the branch of $z^{1/4}$ given by $f(z) = \sqrt[4]{|z|}e^{i\theta/4}$, with $0 < \theta < 2\pi$. (i.e. the branch where all fourth roots have positive real and imaginary parts).

   a. Let γ be the bottom half of the unit circle traversed counterclockwise, paramterized as $\gamma(t) = e^{it}$, $\pi < t < 2\pi$. Sketch $f(\gamma(t))$
   . **(7 points)**

   b. Compute $\int_\gamma f(z)dz$. **(7 points)**

7. Consider the curve γ from $-i$ to i shown below **(14 points)**



Compute

$$\int_\gamma z^2 dz$$

Bonus: **(5 points)**

Let γ be a positively oriented simple closed curve bounding a region R. Show that the area of R is computed by the integral

$$\frac{1}{2\pi i}\int_\gamma \overline{z}$$

Hint: Use Green's theorem.

Scratch Paper

Scratch Paper